NOTES (UNLESS OTHERWISE SPECIFIED):

1. THIS DRAWING SPECIFIES THE REQUIREMENTS FOR A PRINTED WIRING BOARD IN ACCORDANCE WITH SPECIFICATION IPC-6012 CLASS 2 (LATEST REVISION).

2. THE PWB MUST BE LEAD FREE ASSEMBLY PROCESS COMPATIBLE AND MUST BE ABLE TO HANDLE A MINIMUM OF 5 CYCLES AT 260 DEGREES CELSIUS FOR 10 SECONDS.

3. BASE MATERIAL - LAMINATE AND PREPREG SHALL MEET IPC-4101D-26, 83 or 98
   Tg - MUST BE GREATER THAN OR EQUAL TO 150 DEGREES CELSIUS.
   Td - MUST BE GREATER THAN OR EQUAL TO 330 DEGREES CELSIUS.

4. COPPER FOIL WEIGHT - SEE STACKUP DETAIL 'A'

5. CHARACTERISTIC IMPEDANCE - SEE DETAIL 'B'

6. MINIMUM CONDUCTIVE WIDTH/SPACING TO BE .004"/.005"

7. PLATING FINISH: A. BOTH SIDES ENIG: TO MEET THE REQUIREMENTS OF IPC-4552 (LATEST REVISION).

8. FAB VENDOR IS NOT ALLOWED TO USE ODB FOR FABRICATION. CAN BE USED ONLY FOR REFERENCE.

9. SOLDERMASK - TO MEET THE REQUIREMENTS OF IPC-SM-840E (OR LATEST REVISION).
   BLACK COLOR, BOTH SIDES. MODIFICATION OF SOLDERMASK IS NOT ALLOWED WITHOUT WRITTEN PERMISSION FROM NXP.
   TYPE: LPI OR EQUIVALENT.
   A. LOCATION = +/-.002" OF PLATED PADS.
   B. DIAMETER OR SIZE = +/-.002 OF ORIGINAL DATA

10. SILKSCREEN - WHITE EPOXY OR ACRYLIC INK, BOTH SIDES. NO SILKSCREEN ON ANY EXPOSED COPPER FEATURE.

11. ELECTRICAL TEST - 100% IPCD356. PCB FABRICATOR TO PERFORM A NET COMPARE AGAINST THE IPCD356 NETLIST PROVIDED BY NXP.

12. PRINTED WIRING BOARD IS TO BE INDIVIDUALLY BAGGED.

13. DFM CHECK MUST BE RUN ON BOARD DATA BEFORE BUILDING BOARDS, UNLESS PRIOR APPROVAL IS GIVEN IN WRITING BY NXP.

14. TEARDROPS MAY BE ADDED AT THE FAB HOUSE TO ALL SIGNAL LAYERS.

15. TWO SOLDER SAMPLES TO BE PROVIDED.

△16 SUPPLIER MARKINGS - ON SECONDARY SIDE ONLY, WHERE SHOWN.
   - MUST BE UL RECOGNIZED AND MUST HAVE AN ID THAT CONFORMS TO UL94V-0

17. THE PWB WILL BE MARKED AS LEAD FREE BY USE OF AN INK STAMP (P6)

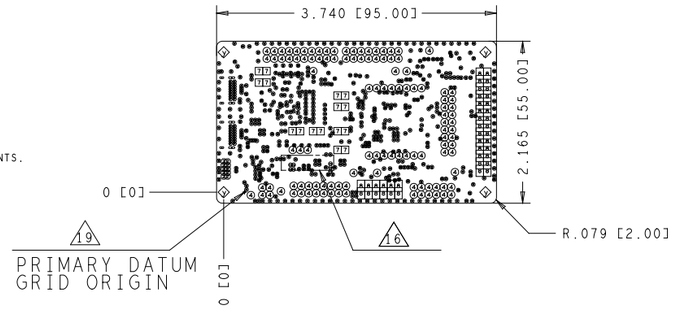
18. THE PWB WILL BE MARKED AS LEAD FREE PROCESS COMPATIBLE BY USE OF AN INK STAMP (260°C)

△19 ALL PLATED AND NON-PLATED THROUGH HOLES ARE TO BE DRILLED AT PRIMARY DRILL STEP. ALL HOLE LOCATION TOLERANCES ARE TO BE +/-.002 IN REFERENCE TO THE PRIMARY DATUM UNLESS OTHERWISE SPECIFIED.

20. FINISHED PCB MUST BE PANELIZED FOR ASSEMBLY ACCORDING TO CONTRACT MANUFACTURERS REQUIREMENTS. THE ADDITION OF RAILS AND .125" NON-PLATED TOOLING HOLES ARE AT THE DISCRETION OF CONTRACT MANUFACTURER. PANELIZATION MUST BE APPROVED BY CONTRACT MANUFACTURER.

21. THIEVING REQUIREMENT:
   COPPER THIEVING ADDITION IS NOT ALLOWED.

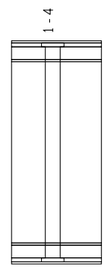
△22 THIS BOARD USES VIA-IN-PAD: SEE FAB_VIAFILL.ART
   A. ALL VIAS USING 10.1 DRILL SIZES ARE TO BE FILLED WITH NON-CONDUCTIVE VIA FILL. LACKWERKE-PETERS PP2795 OR EQUIVALENT AND MADE PLANAR TO THE PADS.
   B. OVERPLATE THE FILLED VIA AND APPLY FINISH METAL TREATMENT.
   C. DIMPLE ON VIA-IN-PADS MUST BE NO GREATER THAN .003" AND PROTRUSION NO GREATER THAN .002"



REVISIONS						
ZONE	REV	DESCRIPTION	DATE	APPROVED DE	APPROVED PE	APPROVED CAD
	A	ORIGINAL RELEASE	09-08-23	Y.Z.	S.D.	O.H.
	A1	CHANGED THE LOCATION OF BH1&BH2	12-29-23	Y.Z.	S.D.	A.Z.
	A2	ADD TEST POINTS (TP5-TP12) ALIGN ARDUINO POSITIONS (J1-J4)	02-26-24	Y.Z.	S.D.	O.H.
	A3	FIX P2_6/LPSPI1_PCS2-ARD_D10 TO P2_6/LPSPI1_PCS1-ARD_D10	04-11-24	Y.Z.	S.D.	O.H.
	A4	UPDATE MCU SYMBOL	06-14-24	Y.Z.	S.D.	O.H.

△22

DRILL CHART: TOP to BOTTOM				
ALL UNITS ARE IN MILS				
FIGURE	FINISHED_SIZE	TOLERANCE_DRILL	PLATED	QTY
⑧	8.0	+0.0/-8.0	PLATED	31
⑩	10.0	+0.0/-10.0	PLATED	502
⑩	10.1	+0.0/-10.1	PLATED	7
⑯	16.0	+2.0/-2.0	PLATED	32
㉘	28.0	+2.0/-2.0	PLATED	10
㊵	40.0	+2.0/-2.0	PLATED	114
41	41.0	+3.0/-3.0	PLATED	40
44	44.0	+2.0/-2.0	PLATED	16
◈	125.0	+2.0/-2.0	NON-PLATED	4
4	55.0x24.0	+2.0/-2.0	PLATED	4
9	83.0x24.0	+2.0/-2.0	PLATED	4



* SURFACE - AIR 0 MIL
* SOLDERMASK_TOP MASK - DRY_FILM_MASK 1.14 MIL
L1 TOP CONDUCTOR - COPPER 2.1 MIL
* DIELECTRIC - S1000HB 3.39 MIL
L2 L2_GND PLANE - COPPER 1.2 MIL

* DIELECTRIC - S1000H 48.43 MIL

L3 L3_PWR PLANE - COPPER 1.2 MIL
* DIELECTRIC - S1000HB 3.39 MIL
L4 BOTTOM CONDUCTOR - COPPER 2.1 MIL
* SOLDERMASK_BOTTOM MASK - DRY_FILM_MASK 1.14 MIL
* SURFACE - AIR 0 MIL

DESIGN CROSS SECTION CHART
TOTAL THICKNESS 64.09 MIL

BOARD THICKNESS TOLERANCE +/-10%
ALL VALUES ARE FINISHED VALUES

DETAIL A
LAYER STACKUP
SCALE: NONE

DETAIL B
IMPEDANCE REQUIREMENTS
IMPEDANCE TOLERANCE IS 10%

LAYERS	SE		DIFF			
	TRACE WIDTH	IMPEDANCE	TRACE WIDTH	TRACE SPACING	IMPEDANCE	REFERENCE LAYER
L1_PS	5.00	50.00	4.00	5.00	90.00	2
L4_SS	5.00	50.00	4.00	5.00	90.00	3

ALL PARTS, MATERIALS AND FINISHED ASSEMBLY SHALL MEET THE RoHS COMMISSION DELEGATED DIRECTIVE (EU) 2015/863 OF 31 MARCH 2015 AMENDING ANNEX II TO DIRECTIVE 2011/65/EU. A CERTIFICATE OF COMPLIANCE IS REQUIRED UPON REQUEST.

ALL PARTS, MATERIALS AND FINISHED ASSEMBLY SHALL NOT CONTAIN ANY OF THE SUBSTANCES OF VERY HIGH CONCERN, (SVHC) ABOVE THE THRESHOLD VALUE PER THE CURRENT ECHA LIST OF SVHC'S AND WITH ANNEX XIV AND ANNEX XVII OF REACH DIRECTIVE 76/769/EEC. A CERTIFICATE OF COMPLIANCE IS REQUIRED UPON REQUEST.

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ARE:
DECIMALS ANGLES
.XX .01 .0-30'
.XXX .005
✓ RMS ALL MACHINED SURFACES
BREAK ALL SHARP EDGES AND CORNERS.
REMOVE BURRS.
UNDERLINED DIM. NOT TO SCALE.
THIRD ANGLE ORTHOGRAPHIC PROJECTION IS USED.

___ COMPANY PUBLIC
X COMPANY INTERNAL
___ COMPANY CONFIDENTIAL

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APPROVALS	DATE
DRAWN OWEN HUANG	06-14-24
CHECKED STEVEN DING	06-14-24
DESIGN ENGINEER YUXING ZOU	06-14-24

PART NO. 170-90841

NXP SEMICONDUCTORS

6501 WILLIAM CANNON DRIVE WEST AUSTIN, TEXAS 78735 USA

TITLE: PRINTED WIRING BOARD
FRDM-MCXA156

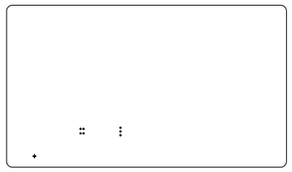
SIZE	CAD FILE NAME	DWG. NO.	REV
D	LAY-90841	FAB-90841	A4

SCALE 1/1 DO NOT SCALE DRAWING SHEET 1 OF 2

NOTES (UNLESS OTHERWISE SPECIFIED):

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	A4	UPDATE MCU SYMBOL	06-14-24	Y.Z.	S.D.	O.H.

22 THIS FAB_VIAFILL.ART SHOWS LOCATIONS OF VIA-IN-PAD TO BE FILLED.



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SIZE	CAD FILE NAME	DWG. NO.	REV
D	LAY-90841	FAB-90841	A4

SCALE 1/1 DO NOT SCALE DRAWING SHEET 2 OF 2

ART FILM - fab viafill